

src/app/graphCityBridge.ts

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Overview

`src/app/graphCityBridge.ts` adds utilities that translate between the knowledge-graph model and the CodeCity layout. It parses graph node IDs, maps them to city building IDs, and generates edge overlays and highlight sets for linked selections.

Key changes

- **Imports** – pulls `GraphEdge`, `KnowledgeGraph` from `@novadiff/graph-core/types` and local layout types (`CodeCityLayoutResult`, `CodeCityRenderableBuilding`, `CodeCityRootSide`) (lines 1-6).
- **CityEdgeOverlay interface** – defines overlay payload (lines 10-14).
- **Node-to-building mapping** – `graphNodeToCityBuildingIds` parses node IDs, matches buildings by kind and symbol, and falls back to file-level matches (lines 74-114).
- **Reverse mapping** – `cityBuildingToGraphNodeId` resolves a building back to the best graph node ID, handling file, function, and class kinds (lines 117-146).
- **Batch mapping** – `buildBuildingIdByGraphNode` pre-computes a map of node IDs to building ID arrays (lines 148-164).
- **Edge overlay generation** – `graphEdgesToCityPairs` filters link-relevant edges, caps results, and deduplicates by source/target pair (lines 167-205).
- **Neighbor discovery** – `graphNeighborNodeIds` returns one-hop neighbors via link edges (lines 208-226).
- **Highlight logic** – `cityBuildingIdsForHighlight` combines node mapping, file-path resolution, and focus-mode neighbor expansion to produce a primary building and a set of highlight IDs (lines 250-307).

Impact

- **Correctness** – new mapping logic aligns graph nodes with city buildings, reducing visual mismatches.
- **Maintainability** – centralizes graph-city translation; future changes to node ID formats or building kinds can be localized.
- **Performance** – pre-computing `buildBuildingIdByGraphNode` and caching neighbor sets keeps per-render overhead low.
- **Compatibility** – no API changes to existing modules; the new file is purely additive.

Risks & follow-ups

- **Edge-case mapping** – verify that `parseGraphNodeId` handles malformed IDs; unit tests should cover all prefixes.

- **Duplicate overlays** – ensure `graphEdgesToCityPairs` ' deduplication logic (`seen` `set`) behaves as intended when multiple edges share the same source/target pair.
- **Focus mode expansion** – confirm that neighbor traversal does not introduce cycles or excessive highlight sets in dense graphs.
- **Type safety** – the new imports rely on `@novadiff/graph-core/types` ; ensure those types are exported correctly and that future version changes do not break the bridge.